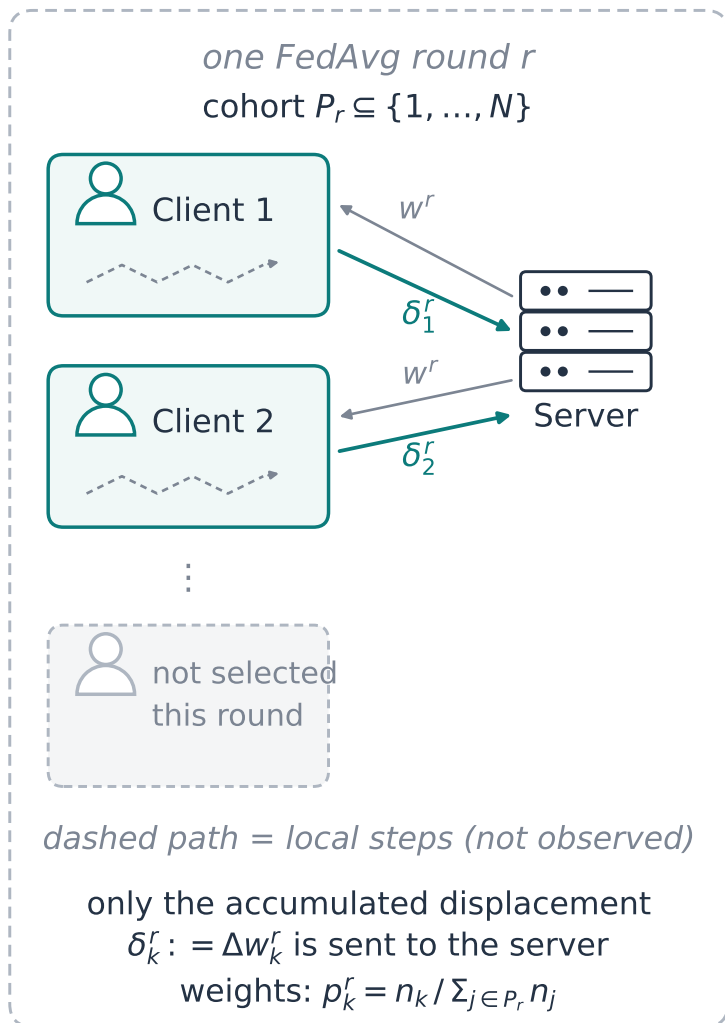


(a) The federated round game (Eq. 5)



Round game u_r (Eq. 5)

coalition $S = \{1, 3\}$: $w^r \xrightarrow{p_1^r \delta_1^r} \xrightarrow{p_3^r \delta_3^r} w^r + \Delta_S^r$

left out $\Rightarrow p_2^r \delta_2^r \mapsto \mathbf{0}$ (zero vector);
kept clients keep their p_k^r (no renormalization)

$u_r(S) = \ell_{\text{val}}(w^r) - \ell_{\text{val}}(w^r + \Delta_S^r)$
= validation-loss decrease from applying only S

grand $S = P_r$: $\bullet \xrightarrow{p_1^r \delta_1^r} \xrightarrow{p_2^r \delta_2^r} \xrightarrow{p_3^r \delta_3^r} \bullet w^{r+1}$

= the actual server update (Eq. 1)

over rounds: $\sum_r u_r(P_r) = \ell_{\text{val}}(w^0) - \ell_{\text{val}}(w^R)$

↓ **2nd-order Taylor** $u_r \rightarrow \hat{u}_r$

Flirds (Eq. 6): exact Shapley value of $\hat{u}_r$

$$\hat{\phi}_k^{(r)} = -p_k^r \langle g_r, \delta_k^r \rangle - \frac{1}{2} p_k^r \langle \delta_k^r, H_r \Delta W_r \rangle$$

$\hat{\phi}_k = \sum_r: k \in P_r \hat{\phi}_k^{(r)}$ (accumulated online per round)
one JVP/round gives $g_r = \nabla \ell_{\text{val}}(w^r)$ and $H_r \Delta W_r$ ($\Delta W_r = \Delta_{P_r}^r$)

(b) What is exact, what is approximated?

Retraining-based Shapley ϕ^{re}

retrain from scratch per subset:
“what if only S had participated?”
the server cannot run it (no raw data)
cost: 2^N retrains



different games, different questions
relation is empirical (§5.2)

In-run Shapley ϕ^{in}

fix the realized trajectory;
enumerate all $2^{|P_r|}$ coalitions
per round; sum over rounds
cost: $\sum_r 2^{|P_r|}$ validation evaluations



↓ **2nd-order Taylor truncation**
the only approximation

Flirds $\hat{\phi}$

the exact Shapley value of the
2nd-order Taylor surrogate $\hat{u}_r$ (Eq. 6)
server-side; no extra client work
cost: 1 JVP + $|P_r|$ inner products / round



both target the round game (Eq. 5)